# Hierarchical Multilayer Distance Fusion kNN for Brain Tumor MRI Classification from Internal CNN Embeddings

Joaquin Mendoza, and Alejandro Veloz

Abstract—Brain tumor MRI classification with convolutional neural networks commonly relies on either the softmax head or the final backbone embedding. This convention is practical, but it treats the last representation as the only downstream view even though internal CNN layers encode a hierarchy of visual abstractions. This paper proposes Hierarchical Multilayer Distance Fusion kNN (HMDF-kNN), a post-hoc method that constructs a compact multilayer distance representation from a fine-tuned CNN without retraining the backbone. Candidate internal embeddings are L2-normalized, evaluated as single-layer metric views on validation data, ranked by validation macro-F1 with balanced-accuracy and accuracy tie-breakers, and combined through validation-selected weighted distance fusion. The selected layer subset, distance weights, and kNN neighborhood size are then frozen before test evaluation. Across 45 brain-tumor MRI dataset-backbone contexts, HMDF-kNN achieved the highest mean test macro-F1 among the evaluated complete methods (0.9616), compared with 0.9550 for multilayer reference methods, 0.9470 for final-embedding classifiers, and 0.9326 for the softmax head. Design ablations supported validation-guided multilayer aggregation over best-single-layer selection (0.9560) and uniform all-layer distance fusion (0.9582), suggesting that the gain is not explained only by using more layers, but by selecting and aggregating complementary layer-wise distance spaces under validation control. Context-level analyses against the strongest validation-selected non-proposed reference method were mostly favorable or practically close under a 0.005 macro-F1 margin. Among 38 contexts with paired prediction artifacts, bootstrap tests with Holm correction identified two significant gains and no significant losses. These results support a bounded aggregate advantage for validation-controlled multilayer distance fusion, rather than a claim of uniform superiority across all dataset-backbone contexts.

Index Terms—Brain tumor MRI, transfer learning, internal embeddings, multilayer fusion, k-nearest neighbors, metric learning, representation learning, Machine-Learning.

## I. Introduction

Brain and central nervous system (CNS) tumors impose a substantial clinical burden because diagnosis, treatment planning, and follow-up depend on imaging evidence that must be interpreted under anatomical, pathological, and acquisition-related variability. Although brain/CNS tumors are less frequent than other cancer types, they are associated with severe neurological consequences and considerable mortality, with GLOBOCAN 2022 reporting more than 320,000 new cases and nearly 250,000 deaths worldwide [1], [2]. Magnetic resonance imaging (MRI) is a core modality for brain tumor assessment because it provides non-invasive anatomical and tissue-dependent information used in diagnosis, follow-up, and treatment planning. However, MRI appearance can vary across tumor subtype, acquisition protocol, contrast sequence, scanner or site conditions, and patient-specific factors, making robust visual classification a non-trivial problem [3], [4].

Deep learning has therefore become a major research direction in brain tumor MRI analysis, including classification, segmentation, multimodal diagnosis, explainability, and representation learning [3], [5], [6]. In many medical-image classification settings, the amount of labeled data is limited relative to the capacity of modern visual architectures. For this reason, practical workflows often adapt pre-trained visual backbones through fine-tuning or feature extraction rather than learning the full representation from scratch [7], [8]. Regardless of the specific backbone, these workflows share a common downstream structure: an image is encoded by a trained model, and the resulting representation is then classified, compared, adapted, or reused [8], [9].

A common and natural choice is to use the final model-derived representation. Standard CNN classifiers typically map the last convolutional stage to a pooled feature vector and then to a learned classification head [10]–[12]. Consequently, once a trained backbone is available, the deepest embedding is often treated as the image descriptor for downstream classification, either through the native softmax head or through a classical classifier applied to the extracted feature vector. Classifiers such as k-nearest neighbors (kNN), support vector machines, random forests, and gradient-boosted trees provide standard downstream alternatives once such an embedding space is available [13]–[16]. This final-embedding strategy is simple, reproducible, and computationally convenient. However, it also imposes a strong representational assumption: the last embedding is treated as the only relevant feature view for the downstream decision rule.

This assumption is not required by the architecture. CNNs produce a hierarchy of internal representations during the same forward pass. Earlier and intermediate layers tend to preserve more local or generic visual structure, such as edges, boundaries, textures, and mid-level patterns, whereas deeper layers become increasingly

J. Mendoza, and A. Veloz are with the School of Biomedical Engineering, Universidad de Valparaiso, Chile. Correspondence: joaquin.mendoza@uv.cl.

specialized for the training objective [7], [17]. Prior work on feature visualization and transferability supports the idea that the semantic and geometric properties of CNN representations vary across depth [7], [17]. Intermediate-representation and multilayer reuse methods further suggest that task-relevant information can be distributed across the network hierarchy rather than being restricted to the final layer [18]–[20]. Thus, the final descriptor is a strong and necessary baseline, but it is not the only representation exposed by a trained CNN.

The central methodological question is therefore not whether internal CNN layers can be used, but how they should be used under a valid evaluation protocol. Selecting a single layer can discard complementary information distributed across depth. Conversely, blindly concatenating or combining all available layers can create a heterogeneous and redundant high-dimensional descriptor. This issue is particularly important for distance-based classifiers, because high-dimensional spaces can reduce the discriminative value of pairwise distances through distance concentration, while redundant or poorly scaled feature blocks can distort neighborhood structure [21]. The opposite extreme is also relevant: overly aggressive compression may remove local, textural, shape, or contrast cues that earlier and intermediate stages preserve [7], [17]. For brain tumor MRI, where subtle appearance differences and acquisition variability can influence class separability, the useful downstream representation is therefore not determined only by network depth; it also depends on how internal views are selected, scaled, compared, and combined.

Multilayer fusion methods differ in where the combination occurs. Architectural fusion combines features during model training or adaptation, often by adding trainable modules, attention mechanisms, or multiscale branches [22]–[24]. Decision-level fusion applies separate classifiers and combines their predictions after classification. Representation-level fusion constructs a shared descriptor before the final decision rule is applied. Distance-level fusion provides a related but distinct formulation: instead of concatenating coordinates into a single vector, it preserves each internal layer as a separate metric view and combines the distances induced by those views. This formulation is attractive in a post-hoc setting because it allows internal representations to contribute to the classifier without retraining the CNN backbone, introducing a new deep fusion module, or constructing a single high-dimensional concatenated descriptor.

This paper proposes Hierarchical Multilayer Distance Fusion kNN (HMDF-kNN), a post-hoc method for constructing a distance-based classifier from internal CNN embeddings under validation control. Given a fine-tuned CNN backbone, the method extracts candidate internal representations from layers at different depths. Spatial activations are converted into vector embeddings through global average pooling (GAP) when necessary, and each resulting layer-specific vector is treated as a candidate internal embedding. Each layer view is then L2-normalized independently. The method evaluates each layer as an individual representation using a distance-weighted kNN criterion on validation data, and ranks layers by validation macro-F1 with balanced-accuracy and accuracy tie-breakers. It then constructs ranked layer prefixes and searches a controlled set of distance-fusion weights and kNN neighborhood sizes using validation data only. After selection, the layer subset, distance weights, and neighborhood size are frozen, and the selected configuration is evaluated once on the test split.

The method treats the internal hierarchy of a trained CNN as a set of candidate metric views and asks whether a compact, validation-selected fusion of layer-wise distances can improve classification over simpler alternatives. This framing is especially relevant in medical-image workflows, where reusing trained feature extractors can be attractive when labeled data, annotation expertise, computational cost, or deployment constraints limit full architecture redesign [3], [7], [8].

HMDF-kNN is evaluated through two complementary analyses. First, internal controls test whether the proposed design is needed. The native softmax head tests whether the fine-tuned CNN classifier is sufficient. Final-embedding classifiers test whether a classical classifier applied only to the deepest available representation is enough. Best-single-layer and uniform all-layer fusion variants then test two simpler explanations for any gain: that performance can be explained by one strong internal layer alone, or that it is sufficient to combine all available layers without validation-guided selection and weighting.

Second, external reference methods test whether established non-proposed mechanisms for reusing or combining embeddings already capture the same benefit. Multilayer and multiview implementations evaluate alternative ways of combining internal CNN views [18]–[20], [25]–[29]. Post-hoc metric and prototype baselines further test whether the observed behavior can be attributed to general embedding-space reuse rather than to the proposed validation-controlled multilayer distance-fusion rule [30]–[32]. This organization is important because a gain from internal embeddings may arise from several different mechanisms, including stronger local geometry, useful intermediate-depth information, reduced dependence on the final layer, multilayer complementarity, or model-selection effects. A fair evaluation must therefore distinguish between internal design evidence and comparison against external alternatives.

The contributions of this paper are:

- a validation-only algorithm for selecting internal CNN layers, distance-fusion weights, and kNN neighborhood size for brain tumor MRI classification;
- a formulation of internal CNN reuse as a validation-controlled multilayer distance-fusion problem, separated from softmax classification, final-embedding classification, best-single-layer selection, blind all-layer fusion, and decision-level voting;
- a design ablation showing that controlled multilayer aggregation improves on best-single-layer selection

and uniform all-layer distance fusion on average;

- a comparison against softmax heads, final-embedding classifiers, all-layer controls, multilayer fusion reference methods, multiview representation methods, and post-hoc metric or prototype baselines under the same train/validation/test protocol;
- an aggregate analysis over 45 brain-MRI dataset-backbone contexts, with HAM10000 reported only as an external-domain control.

## II. Related Work

### A. Brain Tumor MRI Classification and Transfer Learning

Brain tumor MRI classification is commonly addressed as a supervised medical-image classification problem in which an imaging study, slice, or region is assigned to a diagnostic or tumor-related category. Because labeled medical-image datasets are often limited relative to the capacity of modern visual architectures, practical workflows frequently rely on transfer learning, fine-tuning, or frozen-feature reuse rather than learning the full visual representation from scratch [3], [7], [8]. In this setting, a trained backbone may be used directly through its native softmax head, or indirectly as a feature extractor whose embeddings are passed to a downstream classifier. The latter setting is especially relevant for post-hoc evaluation, because it defines the final-embedding baseline: if the deepest representation is already sufficient, a multilayer method must demonstrate value beyond that simpler alternative.

### B. Embedding-Space Classifiers and Distance-Based Reuse

An embedding is a vector representation intended to preserve task-relevant information about an input image. In feature-reuse workflows, embeddings can be passed to downstream classifiers, retrieval procedures, calibration modules, or adaptation methods [8], [9]. Their downstream behavior depends on the dimensionality of the vector space, the scale of its coordinates, and the geometry it induces among samples, that is, which images become close or distant in the resulting feature space.

This geometric view motivates the use of distance-based classifiers. The k-nearest neighbors classifier (kNN) is a transparent downstream method because prediction depends directly on the labels of nearby training samples [13]. Metric-learning approaches, such as Neighborhood Components Analysis (NCA) and Large-Margin Nearest Neighbor learning (LMNN), also emphasize this relation between embedding geometry and neighbor-based prediction by explicitly adapting a representation or metric for local classification [30], [33]. Other classical classifiers, including support vector machines, random forests, and gradient-boosted trees, provide additional final-embedding baselines once a feature vector is available [14]–[16].

Distance-based reuse is sensitive to scale, dimensionality, and redundant coordinates. In high-dimensional spaces, pairwise distances may become less discriminative, a phenomenon commonly associated with distance concentration [21]. This limitation is directly relevant to multilayer CNN embeddings: adding internal representations may introduce complementary information, redundant feature blocks, heterogeneous scales, or poorly aligned distance structures. For this reason, multilayer embedding reuse requires a selection or fusion rule controlled by a valid evaluation procedure.

### C. Internal CNN Representations and Multilayer Feature Fusion

The final embedding of a trained convolutional neural network (CNN) is a natural descriptor because it is directly connected to the original classification head. CNNs also expose multiple internal representations during the same forward pass, and prior work on feature visualization and transferability has shown that these representations vary across network depth [7], [17]. This depth-dependent behavior motivates the use of internal representations as candidate embeddings rather than treating the final vector as the only possible descriptor.

Several works have explored representations from more than one network depth. Fradi et al. extracted pooled convolutional features from multiple CNN layers, normalized and concatenated them, and then applied dimensionality reduction and feature selection for texture classification [18]. Gomes et al. proposed a multilayer feature-fusion method in a few-shot setting, where classification depends on comparing examples in an embedding space [19]. Head2Toe uses intermediate representations for transfer learning by selecting feature groups from multiple depths [20]. In medical-image classification, patch-level, hierarchical, multiscale, and attention-based methods further support the idea that information from different spatial or semantic levels can improve classification when the fusion mechanism is aligned with the task [23], [24], [34]–[36].

These works motivate internal-representation reuse and show that multilayer fusion can be implemented at different stages of the pipeline. Architectural fusion combines features during model training by adding trainable modules, attention blocks, or multiscale branches. Decision-level fusion combines predictions after separate classifiers have been applied. Representation-level fusion constructs a shared descriptor before classification, often through concatenation or projection. Distance-level fusion provides a related but distinct formulation: each layer remains a separate metric view, and the distances induced by selected views are combined before applying a distance-based decision rule. The present work focuses on this last setting.

### D. Multiview, Kernel, and Projection-Based Fusion

Multilayer CNN embeddings can also be interpreted as multiple paired views of the same image, where each view corresponds to a different layer representation. Under this interpretation, multiview representation learning provides relevant reference methods. Generalized canonical correlation analysis (GCCA) seeks a common representation across multiple sets of variables [26], [27]. Generalized multiview analysis (GMA) and multiview discriminant analysis (MvDA) extend this family by incorporating class-discriminative objectives into multiview projection [28], [29]. Multiple kernel learning provides another route by combining kernels defined over different views; EasyMKL is a scalable example of this family [25].

These methods are relevant because they combine several views through projection or kernel weighting. Projection-based multiview methods learn a shared latent representation; kernel-based methods combine view-specific similarity functions. The proposed method follows a different route: it selects a compact set of layer-wise Euclidean distance spaces using validation evidence and performs distance-weighted neighbor voting after freezing the selected configuration. In the experimental comparison, these methods are treated as principled stage-level reference implementations over the same saved CNN embedding views.

### E. Post-Hoc Metric and Prototype Reuse

Post-hoc metric and prototype methods provide complementary forms of representation reuse. NCA followed by kNN tests whether a supervised metric projection over saved embeddings improves local neighborhood classification [30]. Prototype-based methods classify samples by comparing them with class representatives instead of relying directly on all training examples. KMEx fits class-wise prototypes on a trained model's embedding without retraining the model [31], while recent spatial prototype approaches assign image regions to prototypes to provide post-hoc self-explanations [32].

These methods are relevant because they also reuse information extracted from an existing network. Their primary mechanisms differ from multilayer distance fusion. Metric-projection methods modify or project the representation before classification, whereas prototype methods rely on representative class anchors or prototype assignments. The proposed method keeps the selected layer spaces as metric views, combines their layer-wise distances, and applies frozen distance-weighted kNN inference.

### F. Positioning of HMDF-kNN

The preceding literature motivates a post-hoc multilayer classifier evaluated against the native softmax head, final-embedding classifiers, multilayer and multiview reference methods, and metric or prototype-based alternatives under the same train/validation/test protocol. Such a comparison separates possible sources of improvement, including generic view combination, improved embedding geometry, useful intermediate depth, and model-selection effects. Hierarchical Multilayer Distance Fusion kNN (HMDF-kNN) is positioned within this space as a validation-controlled distance-fusion approach: it reuses internal representations from a trained CNN while keeping the backbone architecture fixed, preserving the trained feature extractor during the post-hoc stage, and reserving the test set for final evaluation.

## III. Proposed Method

### A. Overview

This section presents Hierarchical Multilayer Distance Fusion kNN (HMDF-kNN), a post-hoc method for constructing a distance-based classifier from internal CNN embeddings. The method is based on the idea that each internal layer of a trained CNN induces a different metric view of the same image set. Instead of assuming that the final embedding is sufficient, or concatenating all internal representations into a single high-dimensional vector, HMDF-kNN selects a compact set of internal views using validation data and combines their distances before nearest-neighbor voting.

The procedure has five stages, matching the pipeline in Fig. 1. First, a CNN backbone is fine-tuned using the standard supervised training pipeline. Second, after fine-tuning, the backbone is frozen and candidate internal embeddings are extracted for the training, validation, and test splits. Third, candidate layers are scored and ranked on validation data, ranked prefixes are constructed, and candidate fusion rules and neighborhood sizes are searched. Fourth, the selected layer views are combined through weighted multilayer distance fusion. Fifth, the selected layer subset, distance weights, and kNN neighborhood size are frozen before a single final evaluation on the test split.

This design separates representation selection from final test evaluation. The test split is not used to select layers, distance weights, neighborhood size, or any other hyperparameter.

### B. Problem Setup

Let

$$D = D_{\mathrm{tr}} \cup D_{\mathrm{val}} \cup D_{\mathrm{te}} \tag{1}$$

denote the complete dataset partitioned into training, validation, and test splits.

Here, $D_{\mathrm{tr}}$ is used to train the CNN backbone and to provide the reference samples for the post-hoc kNN classifier, $D_{\mathrm{val}}$ is used to select layers, weights, and hyperparameters, and $D_{\mathrm{te}}$ is reserved for final evaluation. The three splits are disjoint.

Each sample is denoted by

$$(x_i, y_i), \tag{2}$$

where $x_i$ is an MRI image and $y_i \in \{1, \ldots, C\}$ is its class label, with $C$ denoting the number of classes in the dataset.

A CNN backbone $f_\theta$ is first fine-tuned on $D_{\mathrm{tr}}$ using the standard supervised training pipeline. During the post-hoc stage, the parameters $\theta$ are kept fixed. Therefore, HMDF-kNN does not retrain the CNN backbone and does not modify its learned feature extractor.

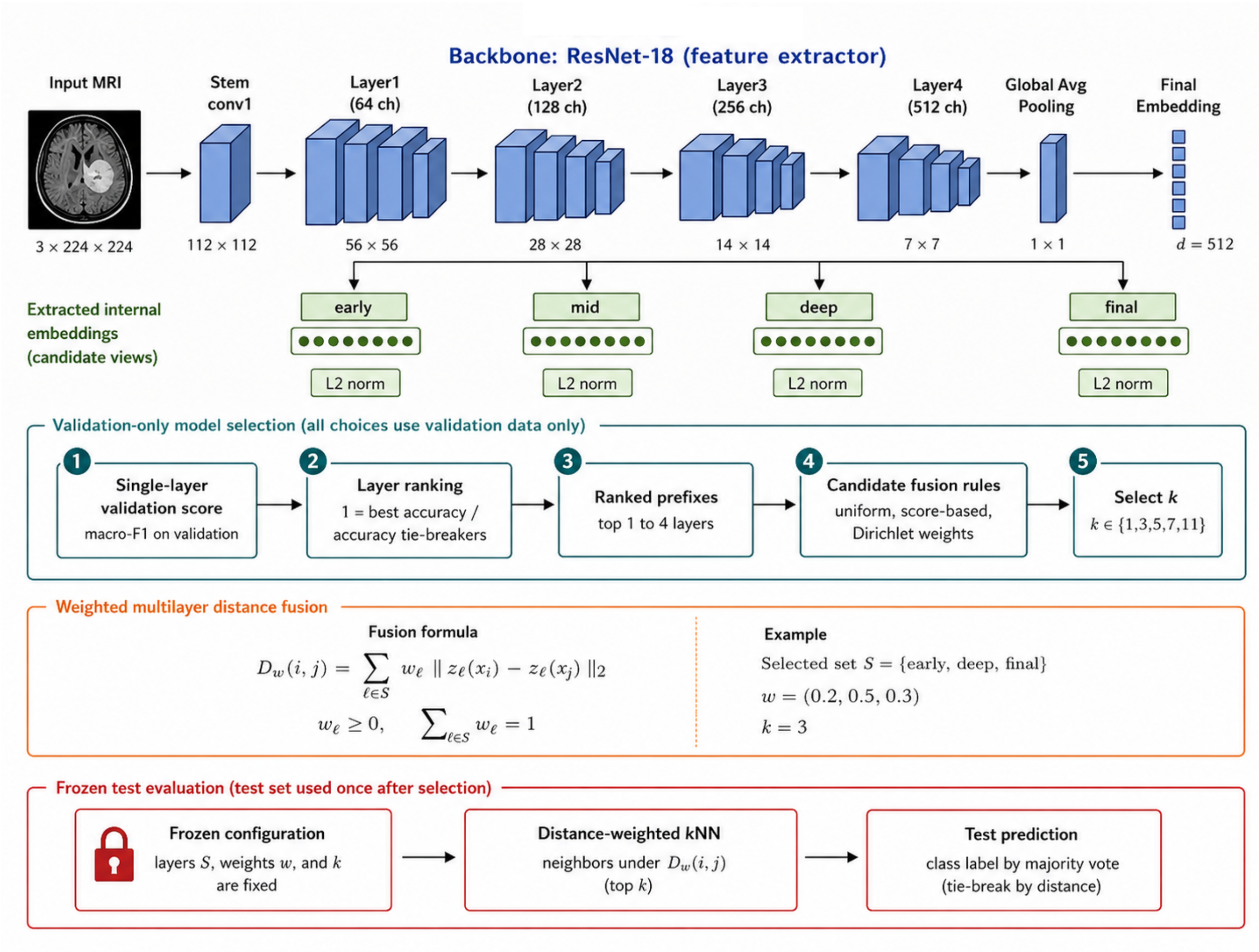


Fig. 1: Overview of HMDF-kNN. A ResNet-18 backbone is shown for illustration as a representative feature extractor; the same post-hoc protocol is applied to all evaluated backbones. The MRI input is passed through the stem, residual stages, global average pooling, and final embedding layer. After supervised fine-tuning, the backbone is frozen and internal representations from early, mid, deep, and final stages are extracted as candidate layer views. Spatial activations are pooled when necessary, and each candidate embedding is L2-normalized independently. Validation-only model selection evaluates each single-layer view with distance-weighted kNN, ranks layers by validation macro-F1 with balanced-accuracy and accuracy tie-breakers, constructs ranked prefixes, searches candidate fusion rules, and selects the neighborhood size $k$. The selected layer subset $S$, distance weights $w$, and neighborhood size $k$ define the weighted multilayer distance $D_w(i,j)$. This frozen configuration is then used for distance-weighted kNN prediction on the test split, which is evaluated only once after model selection.

Let the set of candidate internal layers be

$$\mathcal{L} = \{\ell_1, \ell_2, \ldots, \ell_L\}, \tag{3}$$

where $\mathcal{L}$ is the set of available CNN layers or hooks, $\ell$ denotes one candidate layer, and $L$ is the total number of candidate layers.

For an image $x_i$, layer $\ell$ produces an embedding

$$z_\ell(x_i) \in \mathbb{R}^{d_\ell}, \tag{4}$$

where $z_\ell(x_i)$ is the vector representation of image $x_i$ at layer $\ell$, and $d_\ell$ is the dimensionality of that layer embedding. If the captured layer output is spatial, global average pooling is applied to obtain a vector; if the output is already a vector, it is used directly.

The goal is to construct a classifier from internal embeddings while avoiding two weak extremes. A single top layer may ignore complementary intermediate information, whereas using all layers without control can increase dimensionality, redundancy, and distance instability. HMDF-kNN therefore treats layer selection and distance fusion as validation-controlled model-selection steps.

### C. Layer Views and Normalization

For every candidate layer, train, validation, and test embeddings are extracted from the same frozen backbone. Each layer view is L2-normalized independently:

$$\tilde{z}_\ell(x_i) = \frac{z_\ell(x_i)}{\max(\|z_\ell(x_i)\|_2, \epsilon)}. \tag{5}$$

In Eq. (5), $z_\ell(x_i)$ is the original embedding of image $x_i$ at layer $\ell$, $\tilde{z}_\ell(x_i)$ is the normalized embedding, $\|\cdot\|_2$ is the Euclidean norm, and $\epsilon > 0$ prevents division by zero.

This normalization makes Euclidean distances from different layer views more comparable before fusion. It does not fit a transformation on validation or test data; it only rescales each sample embedding to unit norm.

### D. Validation Ranking of Individual Layers

Each normalized layer is first evaluated as a standalone representation. For every $\ell \in \mathcal{L}$ and each

$$k \in \mathcal{K} = \{1, 3, 5, 7, 11\}, \tag{6}$$

a distance-weighted kNN classifier is fitted on $D_{\mathrm{tr}}$ and evaluated on $D_{\mathrm{val}}$.

For each pair $(\ell, k)$, the validation score is

$$s_{\ell,k} = F1_{\mathrm{macro}}\left(D_{\mathrm{val}}; h_{\ell,k}\right), \tag{7}$$

where $s_{\ell,k}$ is the validation macro-F1 obtained by using layer $\ell$ with neighborhood size $k$, and $h_{\ell,k}$ denotes the corresponding distance-weighted kNN classifier.

For each layer, the best neighborhood size is selected by validation macro-F1, with validation balanced accuracy and validation accuracy used as tie-breakers. If all metrics are tied, the smaller $k$ is preferred. The resulting layer score is

$$s_\ell = s_{\ell,k_\ell^\star}, \tag{8}$$

where $k_\ell^\star$ is the validation-selected neighborhood size for layer $\ell$.

Layers are then ranked by their best validation macro-F1, again using balanced accuracy and accuracy as tie-breakers. The resulting ranking is denoted by

$$\pi = (\pi_1, \pi_2, \ldots, \pi_L), \tag{9}$$

where $\pi_1$ is the highest-ranked layer, $\pi_2$ is the second highest-ranked layer, and so on. This ranking is not interpreted as a clinical explanation of the layer. It is used only to restrict the candidate set for downstream fusion.

### E. Ranked-Prefix Distance Fusion

Given the validation ranking $\pi$, HMDF-kNN constructs ranked layer prefixes:

$$S_m = \{\pi_1, \ldots, \pi_m\}, \quad m = 1, \ldots, \min(R, L). \tag{10}$$

In Eq. (10), $S_m$ is the candidate subset containing the top $m$ validation-ranked layers, $R$ is the maximum prefix length, and $L$ is the total number of candidate layers. In the evaluated implementation, $R = 4$. This keeps the candidate family compact and avoids turning the method into uncontrolled all-layer fusion.

For a selected prefix $S_m$, the fused distance between a query image $x_i$ and a training image $x_j$ is

$$D_w(i,j) = \sum_{\ell \in S_m} w_\ell \left\|\tilde{z}_\ell(x_i) - \tilde{z}_\ell(x_j)\right\|_2. \tag{11}$$

In Eq. (11), $D_w(i,j)$ is the fused distance between images $x_i$ and $x_j$, $w_\ell$ is the non-negative contribution of layer $\ell$, and $\tilde{z}_\ell(x_i)$ and $\tilde{z}_\ell(x_j)$ are the L2-normalized embeddings of the two images at layer $\ell$.

The weights satisfy

$$w_\ell \geq 0, \quad \sum_{\ell \in S_m} w_\ell = 1. \tag{12}$$

Thus, each weight controls the contribution of one internal layer to the final fused distance.

### F. Candidate Distance-Fusion Weights

For each ranked prefix $S_m$, HMDF-kNN evaluates a compact family of candidate distance-fusion weights. The candidate family includes one uniform weighting rule, three deterministic validation-score weighting rules, and eight reproducible Dirichlet-sampled weight vectors.

The uniform rule assigns equal contribution to every selected layer:

$$w_\ell^{\mathrm{uni}} = \frac{1}{|S_m|}, \quad \ell \in S_m, \tag{13}$$

where $|S_m|$ is the number of layers in prefix $S_m$.

The validation-score weighting rules are defined as

$$w_\ell^{(p)} = \frac{(\max(s_\ell, 10^{-6}))^p}{\sum_{r \in S_m}(\max(s_r, 10^{-6}))^p}, \quad p \in \{0.5, 1, 2\}. \tag{14}$$

In Eq. (14), $w_\ell^{(p)}$ is the score-based weight assigned to layer $\ell$, $s_\ell$ is the validation macro-F1 score of layer $\ell$, and the lower bound $10^{-6}$ prevents a zero score from producing a degenerate weight. The exponent $p$ controls how strongly higher-scoring layers are emphasized. Values $p < 1$ produce more balanced weights, while values $p > 1$ assign more emphasis to stronger validation-ranked layers.

In addition to these deterministic weights, HMDF-kNN evaluates reproducible Dirichlet-sampled candidates. For a selected prefix $S_m$, the Dirichlet concentration parameter for layer $\ell$ is computed as

$$\alpha_\ell = \mathrm{clip}\left(\frac{s_\ell^+}{\bar{s}_{S_m}^+}, 0.25, 8.0\right), \quad \ell \in S_m. \tag{15}$$

In Eq. (15), $s_\ell^+ = \max(s_\ell, 10^{-6})$, $\bar{s}_{S_m}^+ = |S_m|^{-1}\sum_{r \in S_m} s_r^+$, and $\alpha_\ell$ is the resulting concentration parameter for layer $\ell$. The function $\mathrm{clip}(\cdot, 0.25, 8.0)$ restricts each concentration parameter to the interval $[0.25, 8.0]$.

For each prefix, eight candidate weight vectors are sampled as

$$w \sim \mathrm{Dirichlet}(\alpha), \tag{16}$$

where $\alpha = (\alpha_\ell)_{\ell \in S_m}$ is the vector of concentration parameters. The sampling uses one NumPy random generator initialized with seed 42 before the prefix search, making the candidate sequence reproducible. Candidate weight vectors that are identical after rounding to eight decimal places are evaluated only once.

The purpose of this candidate family is not to learn a continuous fusion model, but to evaluate a controlled and reproducible set of plausible distance-fusion rules under validation control.

For each prefix and weight vector, the neighborhood size $k$ is again searched over $\mathcal{K} = \{1, 3, 5, 7, 11\}$ using validation data only.

## G. Distance-Weighted kNN Prediction

For a query image $x_i$, let $\mathcal{N}_k^w(x_i)$ denote the set of indices of the $k$ nearest training samples under the fused distance $D_w$. Each neighbor votes with a weight inversely proportional to its fused distance:

$$a_j(x_i) = \frac{1}{\max(D_w(i,j), 10^{-8})}, \quad j \in \mathcal{N}_k^w(x_i). \tag{17}$$

In Eq. (17), $a_j(x_i)$ is the voting weight of training sample $x_j$ for query $x_i$, $D_w(i,j)$ is the fused distance between them, and the lower bound $10^{-8}$ prevents division by zero.

The class vote for class $c$ is

$$V_c(x_i) = \sum_{j \in \mathcal{N}_k^w(x_i)} a_j(x_i)\, \mathbb{I}(y_j = c), \tag{18}$$

where $V_c(x_i)$ is the accumulated vote for class $c$, $y_j$ is the label of training sample $x_j$, and $\mathbb{I}(y_j = c)$ is an indicator function equal to 1 when $y_j = c$ and 0 otherwise.

The predicted class is

$$\hat{y}(x_i) = \arg \max_{c \in \{1,\ldots,C\}} V_c(x_i). \tag{19}$$

This rule preserves the local-neighborhood interpretation of kNN while allowing multiple internal CNN layers to contribute to the distance used for neighbor retrieval.

## H. Validation Selection and Frozen Test Evaluation

Each candidate configuration is defined as

$$M = (S_m, w, k), \tag{20}$$

where $S_m$ is a ranked layer prefix, $w$ is a distance-fusion weight vector, and $k$ is the kNN neighborhood size.

Each candidate is evaluated on $D_{\text{val}}$. The final candidate is selected lexicographically by validation macro-F1, validation balanced accuracy, validation accuracy, and smaller summed layer dimensionality:

$$M^\star = \arg \max_M \left[F1_{\text{macro}}, \text{BalAcc}, \text{Acc}, -\text{Dim}(S_m)\right]_{\text{val}}. \tag{21}$$

In Eq. (21), $M^\star$ is the selected configuration, $F1_{\text{macro}}$ is validation macro-F1, BalAcc is validation balanced accuracy, Acc is validation accuracy, and $\text{Dim}(S_m)$ is the summed dimensionality of the selected layer views:

$$\text{Dim}(S_m) = \sum_{\ell \in S_m} d_\ell. \tag{22}$$

The negative sign in $-\text{Dim}(S_m)$ means that, when validation performance is tied, the structurally smaller configuration is preferred. Because HMDF-kNN fuses distances rather than concatenating embeddings, $\text{Dim}(S_m)$ is used only as a complexity tie-breaker, not as the dimensionality of a final concatenated vector.

After $M^\star$ is selected, the layer subset, distance weights, and neighborhood size are frozen. Test metrics are then computed once on $D_{\text{te}}$. No test labels are used during layer ranking, weight selection, neighborhood-size selection, or any other model-selection step.

## I. Algorithmic Summary

**Algorithm 1** HMDF-kNN validation-controlled multilayer distance fusion

**Require:** Frozen backbone $f_\theta$, candidate layers $\mathcal{L}$, splits $D_{\text{tr}}, D_{\text{val}}, D_{\text{te}}$, neighborhood grid $\mathcal{K}$, maximum prefix length $R$

**Ensure:** Frozen classifier configuration and test predictions

```
Extract embeddings z_ℓ(x) for all ℓ ∈ L and all splits
L2-normalize every layer view independently
for each layer ℓ ∈ L do
    for each k ∈ K do
        Fit distance-weighted kNN on D_tr using layer ℓ
        Evaluate macro-F1, balanced accuracy, and accuracy on D_val
    end for
    Keep the best layer-specific k and validation score s_ℓ
end for
Rank layers by validation macro-F1 with balanced-accuracy and accuracy tie-breakers
Construct ranked prefixes S_m = {π_1, ..., π_m}, for m ≤ min(R, L)
for each ranked prefix S_m do
    for each candidate weight vector w do
        Build D_w using Eq. (11)
        for each k ∈ K do
            Evaluate distance-weighted kNN on D_val
        end for
    end for
end for
Select the best candidate by validation macro-F1, balanced accuracy, accuracy, and smaller summed dimensionality
Freeze selected layers, weights, and k
Evaluate the frozen classifier once on D_te
```

# IV. Experimental Protocol

## A. Datasets and Backbones

The main evaluation uses five brain-tumor MRI datasets and treats HAM10000 as an external-domain control, not as part of the principal brain-MRI averages. Table I summarizes the evaluated datasets. Available metadata support image-level or file-level splits; patient-level separation was not verified. Saved sample-path audits did not detect duplicated sample IDs across train, validation, and test.

Nine CNN backbones were evaluated: ResNet-18, ResNet-34, ResNet-50 [10], DenseNet-121 [11], EfficientNet-B0/B2/B3 [12], MobileNetV3-Large [37], and ConvNeXt-Tiny [38]. Each dataset-backbone pair defines one context. The main brain-MRI analysis therefore contains 45 contexts.

## B. Comparison Methods

The evaluation separates direct CNN baselines, final-embedding classifiers, multilayer controls, literature-inspired multilayer and multiview reference methods, post-hoc prototype references, and HMDF-kNN. Table II summarizes their role.

All post-hoc comparison methods were evaluated on the same saved embeddings, labels, datasets, backbones, and train/validation/test partitions. Transformations such as PCA, discriminant projections, prototype fitting, and kernel weights were fit on train data and selected using validation data. Test metrics were computed only after freezing the selected configuration. Methods originally defined over feature maps or different feature formats were implemented as stage-level adaptations using the available CNN embedding views. Stage-level adaptations, hyperparameter grids, and complete per-context outputs are documented in the supplementary artifacts and will be released with the code repository at [GitHub repository placeholder].

## C. Metrics and Selection Rules

The primary metric is macro-F1 because the evaluated datasets include class imbalance and multi-class settings. Accuracy and balanced accuracy are reported as secondary metrics. AUROC and AUPRC are not used as global ranking metrics because not every method produced calibrated probability outputs. Model selection uses validation macro-F1 with balanced accuracy and accuracy as tie-breakers. The test split is used only after the selected candidate is frozen.

For the context-level comparison, the strongest non-proposed reference method is selected independently in each context using validation macro-F1 and the same validation tie-breakers. Numerical test differences are complemented by a practical equivalence margin of 0.005 macro-F1. Statistical comparisons use 2000 paired, class-stratified bootstrap resamples of the methods' predictions on identical test samples. Per-sample predictions were available for 38 of the 45 selected reference comparisons; two-sided bootstrap probabilities are therefore corrected across those 38 tests using the Holm procedure.

TABLE I: Datasets used in the evaluation.

| Dataset | Role | Classes | Train/val/test | Total | Split |
|---|---|---|---|---|---|
| Brain tumor MRI 15C | Brain MRI main | 15 | 3120/667/669 | 4456 | Image/file-level |
| Brain tumor MRI 17C | Brain MRI main | 17 | 3111/668/669 | 4448 | Image/file-level |
| Brain tumor MRI 44C | Brain MRI main | 44 | 3137/671/671 | 4479 | Image/file-level |
| Brain tumor MRI 4C | Brain MRI main | 4 | 4760/840/1600 | 7200 | Image/file-level |
| ScienceDB brain tumor 3C | Brain MRI main | 3 | 2145/459/460 | 3064 | Image/file-level |
| HAM10000 skin 7C control | External control | 7 | 7010/1502/1503 | 10015 | Image/file-level |

Patient-level separation was not verified from available metadata; no duplicated saved sample IDs were detected across splits.

TABLE II: Compared method families.

| Methods | Group | Role in the comparison |
|---|---|---|
| Softmax head | Direct CNN baseline | Fine-tuned classifier output. |
| Final-embedding selected classifier | Final embedding baseline | Validation-selected classical classifier over the final embedding. |
| Concat, PCA, soft-vote, kernel controls | Multilayer controls | Non-proposed controls for all-layer concatenation, dimensionality reduction, decision fusion, and uniform kernels. |
| MLCFF, Head2Toe, EasyMKL | Literature multilayer references | Stage-level adaptations using the same saved CNN embedding views. |
| MAXVAR-GCCA, GMLDA, MvDA | Multiview references | Internal CNN layers treated as paired views of each image. |
| NCA + kNN; KMEx | Metric/prototype references | Supervised metric projection or final-embedding prototype classification. |
| HMDF-kNN | This work | Validation-ranked weighted fusion of layer-distance matrices followed by frozen kNN evaluation. |

External multilayer methods were adapted at stage level so that all methods used the same frozen CNN embedding views.

# V. Results

## A. Internal Layer Behavior

Before evaluating the complete method, we examined whether internal CNN layers provided useful candidate metric views under the validation protocol. Each layer was evaluated independently with distance-weighted kNN, and validation macro-F1 was used only to characterize and rank the candidate views. Because the backbones expose different numbers of hooks, layer positions were normalized within each architecture and grouped as early, middle, deep, or final.

Figure 2A reports standalone validation performance after first averaging layers within each depth group and context. Mean macro-F1 increased from 0.8920 for early views to 0.9666 for deep views, while final views reached 0.9582. Deep and final representations were strong on average, but early and middle views also provided useful validation signals. Figure 2B shows that these earlier views also appeared in a subset of the validation-selected prefixes. These diagnostics support treating internal layers as candidate metric views; they are not test-set performance claims or clinical explanations.

## B. Selected HMDF-kNN Configurations

Figure 3 characterizes the complete configurations selected by validation. HMDF-kNN selected one layer in 6 contexts, two in 11, three in 15, and four in 13. The neighborhood size was $k = 1$ in 41 contexts. Uniform, deterministic score-based, and finite reproducible Dirichlet candidate weights were selected in 7, 2, and 36 contexts, respectively. These frequencies show that validation controls the number of metric views, the neighborhood size, and the distance-weight rule rather than applying one fixed configuration to every dataset-backbone context. They describe classifier behavior and should not be interpreted as clinical explanations of tumor appearance.

## C. Design Ablation of the HMDF-kNN Aggregation Strategy

Table III studies the internal aggregation design rather than context-level wins or losses. The best validation-ranked single layer reached 0.9560 mean test macro-F1. Uniform all-layer distance fusion increased the mean to 0.9582, indicating that multilayer aggregation can be beneficial. However, blindly aggregating every available layer remained below the validation-ranked prefix variants and the complete HMDF-kNN configuration. Ranked-prefix and greedy constructions reached 0.9599–0.9606, while the complete method reached 0.9616 using an average of 2.78 layer views.

The pattern suggests that the aggregate result is not explained only by adding more layers. Instead, it is consistent with selecting and aggregating complementary layer-wise distance spaces under validation control. Because the variants may jointly change layer selection, aggregation, and $k$, this table should not be interpreted as an isolated causal test of the weight family.

## D. Aggregate Comparison by Method Family

After characterizing the internal behavior and aggregation design of HMDF-kNN, we compared the complete frozen method against external reference families. Figure 4 reports mean test macro-F1 over the 45 brain-MRI

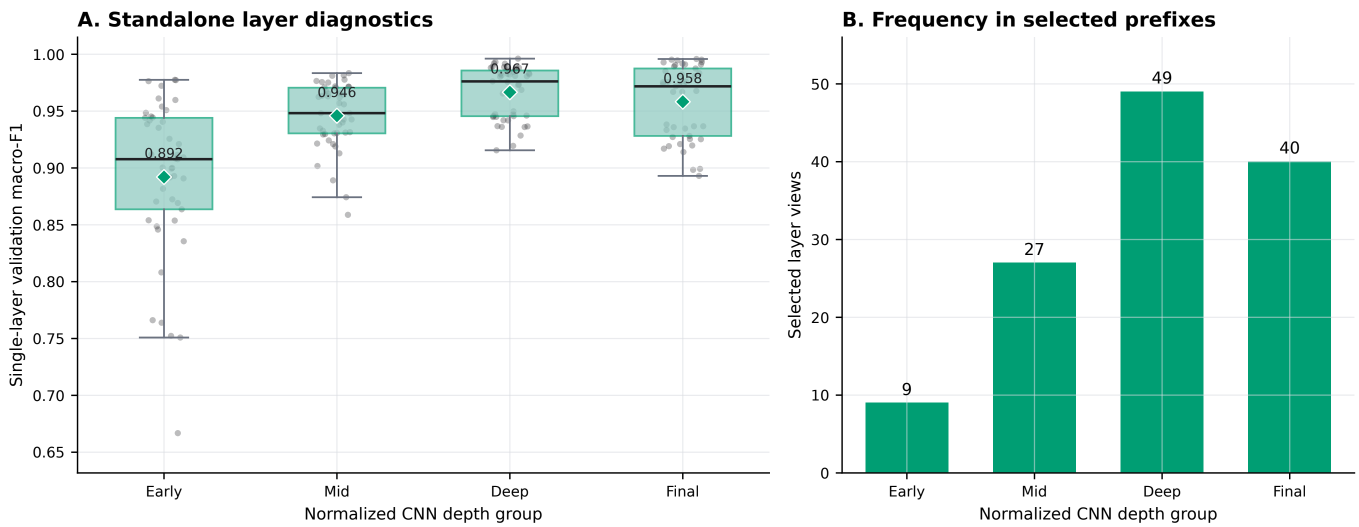


Fig. 2: Internal-layer diagnostics over 45 brain-MRI dataset-backbone contexts. Panel A shows standalone validation macro-F1 by normalized CNN depth group. Layer scores are first averaged within each depth group and context so that architectures contribute equally. Panel B counts the selected layer views appearing in validation-selected HMDF-kNN prefixes. These diagnostics describe validation behavior and are not test-set performance claims.

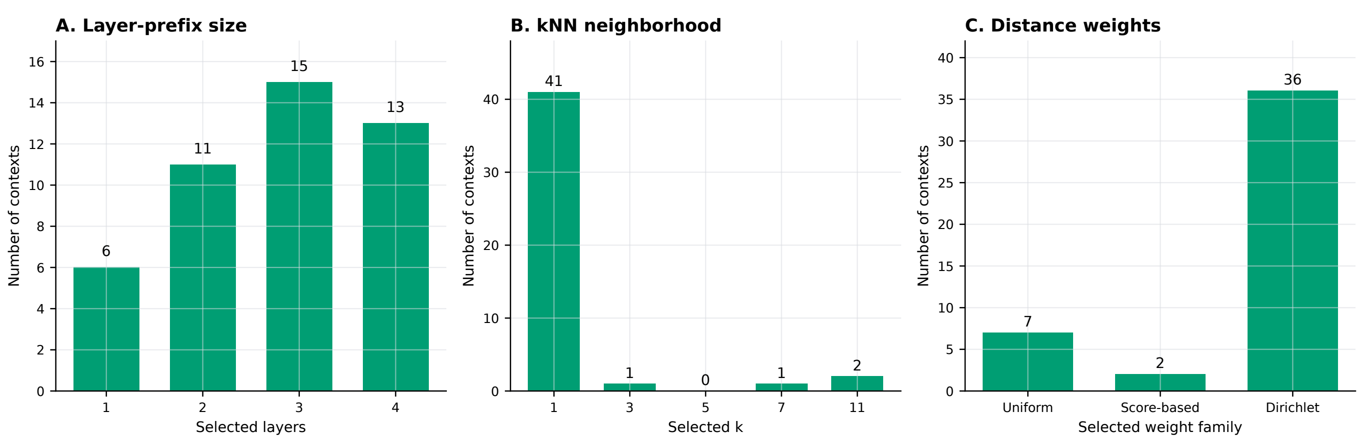


Fig. 3: Validation-selected HMDF-kNN configurations over 45 brain-MRI contexts. Panel A shows the selected ranked-prefix length, Panel B the selected kNN neighborhood size, and Panel C the selected distance-weight family. Score-based weights group deterministic validation-score weighting rules; Dirichlet groups the finite reproducible candidate set evaluated for each prefix.

contexts. The softmax head reached 0.9326, validation-selected final-embedding classifiers reached 0.9470, multilayer reference methods reached 0.9550, and HMDF-kNN reached 0.9616. These values show a favorable aggregate progression toward representation-aware post-hoc methods, but do not imply uniform superiority in every context.

## E. Complete Aggregate Method Comparison

Table IV expands the family-level summary to the specific evaluated methods. The strongest non-proposed aggregate results were obtained by the multiview references MAXVAR-GCCA and MvDA, with mean macro-F1 values of 0.9515 and 0.9504. Among the individual final-embedding classifiers, kNN reached 0.9469. HMDF-kNN obtained the highest listed mean accuracy, macro-F1, and balanced accuracy at 0.9683, 0.9616, and 0.9597, respectively. These are averages across dataset-backbone contexts and should be interpreted as aggregate evidence rather than universal context-level superiority.

Methods whose principal reference is from 2020 onward include author and year directly in the method name for clarity. Stage-level adaptation details, hyperparameter grids, and complete per-context outputs are documented in the supplementary artifacts. The public code location remains a submission task: [GitHub repository placeholder].

TABLE III: Design ablation of HMDF-kNN aggregation strategy.

| Variant | Aggregation strategy | Layer selection | Mean layers | Mean test macro-F1 |
|---|---|---|---|---|
| Best single layer | None | Best validation-ranked layer | 1.00 | 0.9560 |
| All-layer uniform distance fusion | Uniform distance aggregation | All layers | 5.67 | 0.9582 |
| Ranked-prefix uniform fusion | Uniform distance aggregation | Validation-ranked prefix | 2.49 | 0.9599 |
| Ranked-prefix score-weighted fusion | Score-based distance aggregation | Validation-ranked prefix | 2.49 | 0.9599 |
| Greedy score-weighted fusion | Score-based distance aggregation | Greedy selected subset | 2.56 | 0.9604 |
| Greedy uniform fusion | Uniform distance aggregation | Greedy selected subset | 2.60 | 0.9606 |
| HMDF-kNN | Validation-selected weighted distance aggregation | Validation-ranked prefix | 2.78 | 0.9616 |

Results are averaged over 45 brain-MRI contexts. Rows summarize complete validation-selected design variants and may jointly differ in layer selection, aggregation, and $k$; they are not an isolated test of the weight family.

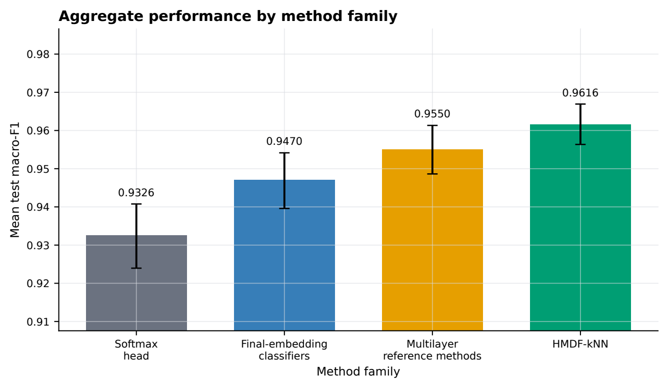


Fig. 4: Mean test macro-F1 by method family over 45 brain-MRI dataset-backbone contexts. Error bars indicate 95% bootstrap confidence intervals across contexts. For each non-proposed family and context, the reported method is selected by validation macro-F1. HAM10000 is excluded.

TABLE IV: Complete aggregate method comparison grouped by method family.

| Method family | Method | $n$ | Acc. | Macro-F1 | Bal. acc. |
|---|---|---|---|---|---|
| Direct CNN baseline | Softmax head | 45 | 0.9386 | 0.9326 | 0.9338 |
| Final-embedding classifiers | Validation-selected final classifier | 45 | 0.9534 | 0.9467 | 0.9499 |
| | kNN | 45 | 0.9537 | 0.9469 | 0.9516 |
| | Linear SVM | 45 | 0.9395 | 0.9323 | 0.9330 |
| | Random forest | 45 | 0.9300 | 0.9219 | 0.9126 |
| | Logistic regression | 45 | 0.9257 | 0.9185 | 0.9259 |
| | Diagonal GMM | 45 | 0.9191 | 0.9148 | 0.9102 |
| | XGBoost | 45 | 0.9199 | 0.9096 | 0.9003 |
| | Gaussian naive Bayes | 45 | 0.9130 | 0.9055 | 0.8894 |
| Multilayer controls | Raw concatenation + linear | 45 | 0.9409 | 0.9347 | 0.9305 |
| | Concatenation + PCA + linear | 45 | 0.9412 | 0.9352 | 0.9310 |
| | Uniform layer soft vote | 45 | 0.9224 | 0.9121 | 0.9005 |
| | Uniform kernel SVM | 45 | 0.9437 | 0.9369 | 0.9341 |
| Literature multilayer references | MLCFF-style fusion (Fradi et al., 2021)$^{\dagger}$ [18] | 45 | 0.9416 | 0.9345 | 0.9314 |
| | Head2Toe-style fusion (Evci et al., 2022)$^{\dagger}$ [20] | 45 | 0.9352 | 0.9282 | 0.9258 |
| | EasyMKL$^{\dagger}$ | 45 | 0.9376 | 0.9307 | 0.9289 |
| Multiview references | MAXVAR-GCCA$^{\dagger}$ | 45 | 0.9584 | 0.9515 | 0.9496 |
| | GMLDA$^{\dagger}$ | 45 | 0.9426 | 0.9360 | 0.9320 |
| | MvDA$^{\dagger}$ | 45 | 0.9573 | 0.9504 | 0.9462 |
| Metric/prototype references | NCA + kNN$^{\dagger}$ | 45 | 0.9486 | 0.9415 | 0.9408 |
| | KMEx (Gautam et al., 2024) [31] | 45 | 0.9383 | 0.9328 | 0.9282 |
| This work | HMDF-kNN | 45 | 0.9683 | 0.9616 | 0.9597 |

Each row averages 45 brain-MRI dataset-backbone contexts. Hyperparameters are selected by validation macro-F1 within each method and context. Individual final-embedding rows select the best hyperparameter setting for that classifier. $^{\dagger}$Stage-level adaptation over the saved CNN embedding views; complete configurations and per-context outputs are provided in the supplementary artifacts.

### F. Context-Level Robustness

Context-level analyses were used as robustness checks rather than as the primary narrative. The aggregate advantage of HMDF-kNN was not uniform across all dataset-backbone contexts; most paired comparisons were favorable or practically close under the 0.005 macro-F1 margin, while only a small number favored the validation-selected reference method. Paired bootstrap differences were statistically resolved in only two contexts after Holm correction, and none significantly favored a reference method among the 38 contexts with aligned prediction artifacts. This supports a cautious aggregate interpretation rather than a claim of universal superiority. Dataset-level summaries, the complete context heatmap, numerical and statistical outcome tables, class-level confusion matrices, and the qualitative fused-distance visualization are provided as supplementary artifacts.

External-domain control: HAM10000 was evaluated only to stress-test the post-hoc procedure outside brain MRI. It is excluded from brain-MRI averages, context counts, and inferential tests; the main claims remain restricted to the five brain-MRI datasets.

## VI. Discussion

The results support a bounded interpretation of HMDF-kNN: validation-controlled fusion of internal CNN distance spaces can improve average brain-MRI classification performance under the evaluated split protocol. The complete method achieved the highest mean test macro-F1 among the evaluated complete methods, while strong references such as MAXVAR-GCCA and MvDA remained competitive. This indicates that the observed gain should not be interpreted as a generic advantage of any internal-layer reuse strategy, but as evidence for the specific procedure evaluated here: validation-ranked layer selection, weighted distance fusion, and frozen distance-weighted kNN inference.

The internal analyses clarify why the method is useful. Single-layer diagnostics showed that deep and final representations were strong on average, but earlier and middle layers also provided validation signal and were retained in some selected prefixes. The ablation further showed that using only the best validation-ranked layer was not sufficient on average, while uniform fusion of all available layers also remained below the complete method. This pattern suggests that the benefit is not explained simply by using more layers. Rather, it is consistent with the idea that complementary layer-wise metric views must be selected and aggregated under validation control.

The selected configurations also show that HMDF-kNN behaves as a compact post-hoc model rather than as uncontrolled all-layer fusion. Across contexts, the method selected a small ranked prefix of layer views and used the selected distance weights only after freezing the validation-selected configuration. The frequent selection of very local neighborhoods suggests that the fused metric often produced useful nearest-neighbor structure, although this behavior should be tested further under additional seeds and external validation settings.

The comparative results should also be read cautiously. Context-level analyses showed that the aggregate advantage was not uniform across all dataset-backbone contexts. Most paired comparisons were favorable or practically close under the predefined margin, but only a small number of paired differences remained statistically significant after Holm correction. This supports an aggregate performance claim, not a claim of universal superiority over every reference method in every context.

Several limitations remain. First, patient-level separation was not verified from the available metadata, so the reported splits should be interpreted as image-level or file-level splits. Second, the main results are based on the available trained backbones and seed, and multi-seed evaluation is needed before making stronger stability claims. Third, the study does not establish clinical deployment readiness, external clinical validity, or biomarker-level interpretability of selected layers. Fourth, computational cost was not measured through latency, FLOPs, or memory usage; compactness was assessed only through the number of selected layer views. Finally, several literature methods were implemented as stage-level adaptations over the saved CNN embedding views, so they should be interpreted as principled reference implementations rather than full reproductions of the original pipelines.

Future work should therefore prioritize patient-level or institution-level validation when metadata permit it, multi-seed robustness analysis, measured computational efficiency, and calibration analysis for methods with reliable probability outputs. It would also be useful to evaluate whether the selected layer prefixes and distance-weight patterns remain stable across external MRI cohorts, acquisition protocols, and larger backbone families.

## VII. Conclusion

This work presented Hierarchical Multilayer Distance Fusion kNN (HMDF-kNN), a post-hoc method for brain tumor MRI classification that reuses internal CNN embeddings as layer-wise metric views. Instead of relying only on the softmax head or the final backbone embedding, HMDF-kNN ranks candidate internal layers on validation data, selects a compact multilayer prefix, fuses the corresponding distance spaces, and freezes the selected configuration before test evaluation. Across 45 brain-MRI dataset-backbone contexts, the method achieved the highest mean test macro-F1 among the evaluated complete methods. The design ablation indicates that validation-controlled multilayer aggregation improves on both best-single-layer selection and uniform all-layer distance fusion on average, suggesting that the main contribution is not simply the use of additional layers but the controlled selection and aggregation of complementary internal views. Context-level analyses support a cautious aggregate interpretation: the method was often favorable or practically close to the strongest validation-selected non-proposed reference, but corrected statistical significance was limited. Future work should extend the evaluation to multi-seed experiments, patient-level or external institutional validation, measured computational cost, and calibration analysis when probability outputs are available.

## References


[1] International Agency for Research on Cancer, "Global Cancer Observatory: Brain, central nervous system fact sheet," https://gco.iarc.who.int/media/globocan/factsheets/cancers/31-brain-central-nervous-system-fact-sheet.pdf, 2022, gLOBOCAN 2022.

[2] ——, "Global Cancer Observatory: Chile fact sheet," https://gco.iarc.who.int/media/globocan/factsheets/populations/152-chile-fact-sheet.pdf, 2022, gLOBOCAN 2022.

[3] Dorfner et al., "A review of deep learning for brain tumor analysis in mri," 2025.

[4] "Harmonization strategies in multicenter mri-based radiomics," 2024.

[5] S. Ketabi, M. W. Wagner, C. Hawkins, U. Tabori, B. B. Ertl-Wagner, and F. Khalvati, "Multimodal contrastive learning for enhanced explainability in pediatric brain tumor molecular diagnosis," p. 10943, 2025.

[6] "Explainable deep learning models in medical image analysis," 2024.

[7] J. Yosinski, J. Clune, Y. Bengio, and H. Lipson, "How transferable are features in deep neural networks?" in Advances in Neural Information Processing Systems, 2014.

[8] A. Bär, N. Houlsby, M. Dehghani, and M. Kumar, "Frozen feature augmentation for few-shot image classification," 2024.

[9] X. Li et al., "From embeddings to accuracy: Comparing foundation models for radiographic classification," 2025.

[10] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016.

[11] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017.

[12] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in Proceedings of the International Conference on Machine Learning, 2019.

[13] T. Cover and P. Hart, "Nearest neighbor pattern classification," IEEE Transactions on Information Theory, vol. 13, no. 1, pp. 21–27, 1967.

[14] C. Cortes and V. Vapnik, "Support-vector networks," Machine Learning, vol. 20, pp. 273–297, 1995.

[15] L. Breiman, "Random forests," Machine Learning, vol. 45, pp. 5–32, 2001.

[16] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016.

[17] M. D. Zeiler and R. Fergus, "Visualizing and understanding convolutional networks," in European Conference on Computer Vision, 2014, pp. 818–833.

[18] H. Fradi, A. Fradi, and J.-L. Dugelay, "Multi-layer feature fusion and selection from convolutional neural networks for texture classification," in Proceedings of VISAPP, 2021, pp. 574–581.

[19] J. C. Gomes, L. d. A. B. Borges, and D. L. Borges, "A multi-layer feature fusion method for few-shot image classification," Sensors, 2023.

[20] U. Evci et al., "Head2Toe: Utilizing intermediate representations for better transfer learning," https://github.com/google-research/head2toe, 2022.

[21] K. Beyer, J. Goldstein, R. Ramakrishnan, and U. Shaft, "When is nearest neighbor meaningful?" in International Conference on Database Theory, 1997.

[22] Y. Dai, F. Gieseke, S. Oehmcke, Y. Wu, and K. Barnard, "Attentional feature fusion," 2020.

[23] He et al., "HEMF: An adaptive hierarchical enhanced multi-attention feature fusion framework for cross-scale medical image classification," 2025.

[24] Zhang et al., "Enhancing medical image classification with context modulated attention and multi-scale feature fusion," 2025.

[25] F. Aiolli and M. Donini, "EasyMKL: A scalable multiple kernel learning algorithm," Neurocomputing, vol. 169, pp. 215–224, 2015.

[26] J. D. Carroll, "A generalization of canonical correlation analysis to three or more sets of variables," in Proceedings of the 76th Annual Convention of the American Psychological Association, 1968, pp. 227–228.

[27] J. R. Kettenring, "Canonical analysis of several sets of variables," Biometrika, vol. 58, no. 3, pp. 433–451, 1971.

[28] A. Sharma, A. Kumar, H. Daume, and D. W. Jacobs, "Generalized multiview analysis: A discriminative approach to dimension reduction," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2012, pp. 2160–2167.

[29] M. Kan, S. Shan, H. Zhang, S. Lao, and X. Chen, "Multi-view discriminant analysis," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 38, no. 1, pp. 188–194, 2016.

[30] J. Goldberger, S. Roweis, G. Hinton, and R. Salakhutdinov, "Neighbourhood components analysis," in Advances in Neural Information Processing Systems, 2004.

[31] S. Gautam, A. Boubekki, M. Hoehne, and M. Kampffmeyer, "Prototypical self-explainable models without re-training," https://openreview.net/forum?id=HU5DOUp6Sa, 2024.

[32] A. Boubekki and L. H. Clemmensen, "Post-hoc self-explanation of CNNs," https://openreview.net/forum?id=cKw94d39J0, 2026.

[33] K. Q. Weinberger and L. K. Saul, "Distance metric learning for large margin nearest neighbor classification," Journal of Machine Learning Research, vol. 10, pp. 207–244, 2009.

[34] T. Muezzinoglu et al., "PatchResNet: Multiple patch division-based deep feature fusion framework for brain tumor classification," Journal of Digital Imaging, vol. 36, pp. 973–987, 2023.

[35] Ullah and Kim, "Hierarchical deep feature fusion and ensemble learning for enhanced brain tumor mri classification," 2025.

[36] Sankari et al., "Hierarchical multi-scale vision transformer model for accurate detection and classification of brain tumors," 2025.

[37] A. Howard, M. Sandler, G. Chu, L.-C. Chen, B. Chen, M. Tan, W. Wang, Y. Zhu, R. Pang, V. Vasudevan, Q. V. Le, and H. Adam, "Searching for MobileNetV3," in Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 1314–1324.

[38] Z. Liu et al., "A convnet for the 2020s," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022.